Design and Estimation of Reliability of an Off Grid Solar Photovoltaic (PV) Power System in South East Queensland

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Abstract - **The threat of the changing climate conditions has pushed the world's interest in the renewable energy systems. Solar energy is considered as an ideal renewable energy since it is considered as more cleaner and sustainable sources of energy. However, higher capital cost of installation of 'off - grid solar photovoltaic (PV) system and lack of perception of system reliability are the major drawbacks of the solar systems which can also have a negative effect on the manufacturer's reputation and in some cases, on the technology. Hence the reliability estimation of the solar PV system becomes extremely important. The aim of this research paper is to design and estimate the reliability of an off grid solar PV system. The greatest challenge for the PV system reliability estimation is the changing input to the system which is solar radiation. This research study designs a solar PV system for a house located South east Queensland (Australia) and later estimates the reliability of this designed system. The study used both mathematical and software tools while designing and reliability estimation of the off grid solar PV system. The result of this research can be used to model any other Off grid solar PV system's reliability and availability with some modifications based on climatic factors.**

Keywords - **Photovoltaic (PV) Power System, Solar Energy, Reliability, Weibull Distribution, Reliability Block Diagrams (RBD).Introduction (*Heading 1*)**

I. INTRODUCTION

First, The world is experiencing shortfall of the sources that are used to generate the electricity due to the rapid increase in the electricity consumption. Most of power or electricity around the world is still being derived using non-sustainable fossil fuels such as coal, petrol, diesel and natural gas. Extensive usage of fossil fuel in almost all human activities including power generation led to some unexpected consequences such as environmental pollutions, global warming, greenhouse effect, climate change etc. [1, 2]. The increasing impact of global warming and its negative impacts on the climate around us have forced the mankind to shift its attention towards more cleaner and sustainable sources of energy such as solar energy rather than only using the non-renewable fossil fuelled energy. This sustainable solar energy can be tapped from the sun by using many different techniques. Solar Photovoltaic or solar PV system is one of the most popular and widely used techniques. Australia has the highest average solar radiation per square meter. The Renewable Energy Target (RET) scheme of the Australian government is designed to deliver on the Australian government's commitment to ensure that the equivalent of at least 20 per cent of Australia's electricity will come from renewable sources by 2020 (Dept. of Environment: Australia 20133). The data released by the Australian government showed that more than a million households in Australia are presently equipped with the solar photovoltaic panels [4]. Solar PV systems can be generally divided into two types such as stand-alone PV systems (Off-grid) and grid-connected PV systems (On-grid). The stand-alone PV power system can supply power to the load without connected to the conventional large electrical power supply system whereas, the Grid connected or 'On grid' system is integrated into the nation's distribution network.

Since solar energy is an inexhaustible resource for energy production and more and more households are getting dependent on the solar PV system for their energy requirements, there is an urgent need to check the reliability of the PV systems in order to keep track of their performance and design new systems with the increased efficiency. Reliability of such energy system is also important as lack of or poor reliability can have negative impacts on the manufacturer's reputation as well as the perception of the technology, and most importantly on the users satisfaction. One of the biggest problems in estimating reliability of solar system is that the system's input is not constant as it depends on the variation of intensity of solar light throughout the day. To achieve the aim, this research study designs a solar PV system for a house located South east Queensland (Australia) and later estimates the reliability of this designed system.

II. A BRIEF OVERVIEW OF STAND ALONE PV SYSTEMS

A. Abbreviations Stand - Alone PV Systems:

Define The Off-Grid or Stand-Alone PV System incorporates large amounts of battery storage to provide power for a certain number of days (and nights) in a row when sun is not available. The array of solar panels must be large enough to power all energy needs at the site and recharge the batteries at the same time. Most off - grid systems benefit from the installation of more than one renewable energy generator and may include Wind and/or Hydro power. A gas or diesel generator is often employed for emergency backup power.

III. A BRIEF OVERVIEW OF RELIABILITY

Reliability can be defined as the probability of a system performing adequately for the period of time under the operating conditions encountered [4, 5]. So, it is the characteristic of any entity expressed by probability that it will perform or able to perform a required function under a stated condition for a stated period [6]. The reliability of a solar energy system is important because it can have a detrimental impact on the price and the cost to the manufacturer due to system failure, warranty claims for repairs and finally loss of goodwill. In some cases the safety of the systems or components can be influenced by the reliability [7]. Reliability is also important for solar energy systems due to the fact that they are exposed to the elements of nature. Weather conditions, variation in temperature from one extreme to another on certain area of the globe and other factors play vital roles in determining the reliability of this system. The reliability of a component with density function *f(t)* at any time t, can be expressed as.

$$R(t) = 1 - \int_0^t f(t)dt \qquad (1)$$

To find the reliability of the individual components of a system, testing must be done to find out the failure rate of the components. The type of stresses applied depends on the type of components. The failure rate is adjusted using an acceleration factor which is found by the ratio of life under standard environmental and life under a stress environment [7-9]. The failure rate of a component is the number of failures per unit of time and can be expressed as

$$h(t) = f(t)/R(t) \qquad (2)$$

For constant failure rate, $\lambda = f/\sum t$ (3)

f = Number of Failures during testing time

t = Testing time

λ = Constant failure rate

A. Reliability Block Diagrams

The reliability models are used to predict the expected reliability of a system under test by using the component failure rates. One of the Reliability modeling techniques is Block Diagrams. Here we discuss most commonly used reliability block diagrams.

Series Combination: Connecting the blocks in series is one of the simplest configurations. In the series configuration, if all the blocks are working then the series branch has not failed otherwise as soon as one of the blocks in the series configuration fails; the whole series branch also fail. If a system has N components represented by blocks as A and B,..,& N in series and the reliabilities of each block is represented by RA and RB,…, RN then the overall system reliability would be:

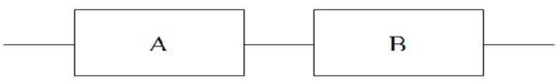

Figure 1: Series Branch Reliability Block Diagram (RBD)

$$R_{System}(N) = R_A \times R_B \times \ldots \times R_N$$

The system could also be represented with the help of constant failure rate as:

$$R_A = e^{-\lambda_A t}$$
$$R_B = e^{-\lambda_B t}$$

The reliability of the system with N components becomes:

$$R_{Sys} = e^{-(\lambda_A + \lambda_B + \cdots \ldots + \lambda_N)t}$$

Parallel Combination: In the parallel system configuration, unlike series combination the failing of one component does not fail the whole system. In this configuration, only a minimum of one block operating prevents the whole system branch from failing [10].

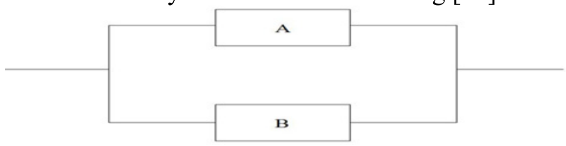

Figure 2: Parallel Branch Reliability Block Diagram

The derivation of the reliability block diagram models are fairly simple but can be more difficult and more time consuming with complex systems [11].

IV. RELIABILITY ESTIMATION OF A STAND ALONE PV SYSTEMS

A. System Design

Since this research estimates the reliability of a stand-alone PV system, the researcher in this research has designed a practical Off Grid Solar PV system for a house located in Southeast Queensland (Australia). The reason behind this selection is that southeast Queensland considered as ideal location for household solar PV system installation since it receives more than 300 days of sunshine annually. While designing an off grid solar PV system, one needs to consider many factors such as weather conditions, altitude angle and selection of components in the system etc. The research used the RetScreen Clean Energy Project Analysis software to estimate various parameters required for designing a system at a particular location. The estimated load of the house was approximately 3 KW. Hence eighteen 175 Watt solar panels have been used for the designed solar PV off grid system. This single crystal 175 Watt module feature 16.4 % encapsulated cell efficiency and 14.2 % module efficiency. The inverters are one of the most vulnerable components in a PV system and require the investment three to five times during the life span of a typical solar PV system. The Inverters were believed to work adequately over the warranty period of 5 years [10]. A Sine Wave Inverter has been used for the house in the designed solar PV system. This Inverter offers a maximum efficiency of 95 % with the wide input range. The inverter comes with the comprehensive warranty.

In an off grid solar PV system the generated DC power from the solar panels is fed to the charge controller which provides a regulated DC output and simultaneously stores the excess charge in the battery banks. The life of a battery can be extended by maintaining its charge and discharge conditions under control in order to prevent battery overcharge and deep discharge conditions. The Aerl Coolmax SR Maximizer (SRMVW) has been used in the designed system. This controller has high efficiency. The COOLMAX SR employs a maximum power point tracking strategy which has been proven to be highly robust, resistant to local extremes, and results in power losses of less than 0.5% over the whole operating temperature range of a PV array. This charge controller also monitors battery voltage to prevent the under charging and the overcharging conditions. The output from the charge controller is further fed as input to the inverter. The Battery plays a very important role in an Off – grid solar PV system. The solar Off - grid system being only dependent on the sunshine for its operation requires the battery bank to store enough charge to meet the energy requirements of a household in the event of no sunshine days. The battery bank in a solar Off - grid system is designed such a way that it could feed the house load for at least a couple of days. If these conditions are not satisfied while designing a system, then the overall reliability of the system is decreased.

Queensland being the state which receives more than 300 days of sunshine is ideal for solar PV systems. Considering the fact that there could be no sunshine for a day or two, the battery banks in this off grid solar PV system are designed to feed the household load for approximately two days. The battery banks play a vital role in estimating the reliability of a solar PV system. Hence to increase the reliability of a stand-alone PV system, one needs to keep an eye on the efficiency of the battery banks installed in the system because the efficiency of the battery banks reduces with its use. The new battery bank might have an efficiency of 90-95 % but after a couple of year's use, the efficiency of the battery bank will drop. Historical data shows that the selected house has the energy requirements of 3000 Watts per day and the system voltage is set to 48 volts then the size of the battery bank is decided with the help of simple mathematics as 3000 / 48 = 62.5 Amp hours per day and If there is no sunshine at the location of the project for 3 days (days of autonomy), then the battery bank specificationnwould be : 2.5 Amp hours x 3 = 187.5 Ah @ 48 Volts. The designed system will have some losses as well. Therefore, by considering the losses in the designed system, it is recommended a battery bank of 240 Ah @ 48 Volts for this designed solar PV power plant. For this purpose, Sonnenschein 48 V 240 Ah (24*2 Volt 240 Ah) solar series battery bank is selected. The Sonnenschein A 600 solar series batteries are developed for medium to large solar powered applications. The recycle ability and the long storage life without recharge make these batteries environmentally friendly solar battery system. The solar tracking mode was fixed. The slope value of 20 was selected and the Azimuth was selected as 180 for maximum efficiency of the system. For estimating the reliability of the designed system ReliaSoft.

B. Reliability Estimation of a Stand Alone PV Systems using Reliability Block Diagrams (RBD):

Headings, The ReliaSoft BlockSim software is used in estimating the reliability of the system or a component present in the system. Due to the absence of the actual field test data, the failure rates and the expected life of the components are assumed to estimate the system reliability using the BlockSim 9 software. To check the overall reliability of this Off grid system using the BlockSim software, one needs to input the parameters for the each component in the system so that the software could compute the reliability of the whole system. While checking the reliability of the system using the RBD, the components of the system are represented in the form of the individual independent blocks. The Figure 3 shows the individual components of an off grid solar PV system in the form of the blocks designed.

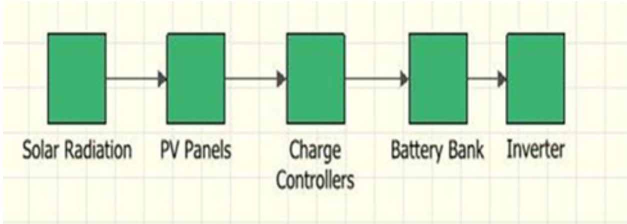

Figure 3: Reliability Block Diagram (RBD) (Source: ReliaSoft BlockSim 9 Software)

The overall reliability is then calculated by the software using the reliability data of individual component. Reliability model option is selected in the software. The Reliability model creates a probabilistic model that would be able to estimate the reliability of the system. In the next step one needs to select the condition related to the components in the system. There could be three conditions for a component in the system as:

- It is more likely that the older item will fail first
- It is more likely that the new item will fail first
- It is equally likely that either the new item or the old item will fail first.

For model simplification purpose it is assumed the older item in the system will fail first. In the next step one needs to input the parameters related to the typical life of the components used in this solar power generation system. Since the solar PV panels selected for the designed system come with twenty five years warranty from the manufacturers of the panels, the data entered in the Quick The design life of the solar PV panels is assumed to be thirty years (Approx. 10800 Days). The same process is repeated for all other components used in the system. After entering the data related to the warranty period and the design life of the Solar PV panels, one needs to enter the failure data for the components in the next step in order to find the reliability of the system. The ReliaSoft BlockSim 9 software calculates the values of β and η on the basis of data entered by the user for each individual component of the system. Hence after entering the data, the system calculates the values of β and η for the solar PV panels. Similar to the solar PV panels, the data has to be entered in the system for each individual component for the whole off grid solar PV system. The table below shows the data entered in the system for various components of the PV system to calculate the overall system reliability. The data related to the warranty period is taken same as the warranty period of the actual components offered by the respective manufacturers of the components. As an example for the solar PV panels, the warranty offered by the manufacturer is twenty five years, in the case of charge controller it is two years and in the case of battery bank it is five years. The table 1 shows the assumed values of the Design Life (Days) and Warranty Period (Days) and the expected failures (% age) at the end of warranty period for different components of the designed system. The table 1 also shows the expected failures (% age) at the end of the design life and the table 1 also shows the values of Beta and Eta calculated by the ReliaSoft BlockSim 9 software based on the values inputted in the software.

The Figure 4 shows the Mean Life of the off grid PV system calculated by the ReliaSoft BlockSim software. This Mean Life is based on the assumed and imputed values of each component of the off grid solar PV system. The value of failure rate is calculated in Days. The Reliable Life of the designed off grid solar PV system is calculated with the required reliability of 0.9. The Figure 5 shows the reliability of the PV system at Time, t = 1 Day.

TABLE 1: EXPECTED FAILURES AT THE END OF WARRANTY PERIOD

Name of the Component	Design Life (Days)	Warranty Period (Days)	Expected Failures (% age) at the End of Warranty Period			Estimate d β value	Estimated η value
			The Worst – Case (Highest) Estimate	The Most Likely Estimate	The Best – Case (Lowest) Estimate		
Solar Radiation	36000	30000	5	3	1	1.606	263741
Solar PV Panels	10800	9000	5	3	1	6.807	15031
Charge Controller	3600	720	8	4	1	0.794	38363
Battery Bank	3600	1800	6	4	1	1.4557	16690
Inverter	3600	1080	8	4	1	0.7673	66068

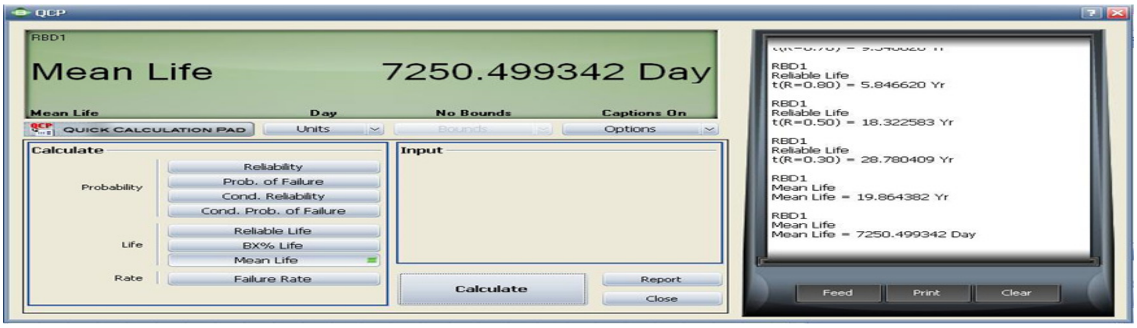

Figure 4: Mean life of the System

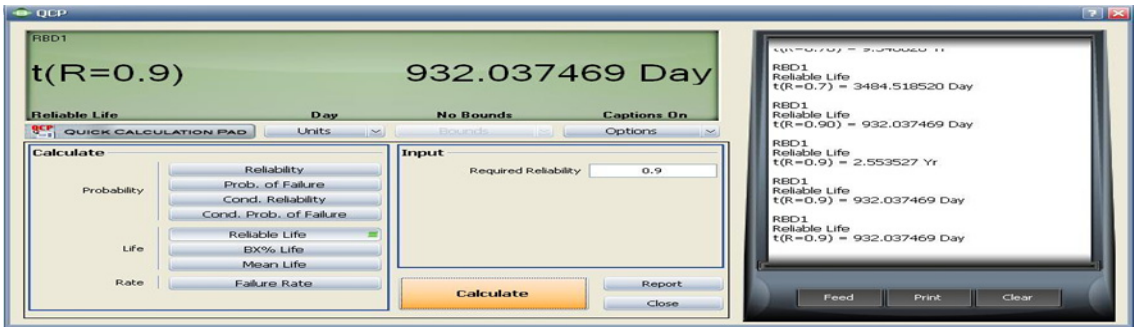

Figure 5: Reliable Life of System at Required Reliability value of 0.9

The Figure 6 shows the Reliability Vs Time graph (at Resolution = 50 Days) for the system. It can be seen from the graph that the reliability of the system is decreasing with time. The Unreliability Vs Time graph (at Resolution = 50 Days) (not shown in the paper because of page limitation) for the system exhibits that the unreliability of the system is increasing with time. Figure 11exhibits the Block Unreliability vs. Time graph at 50 Days resolution and

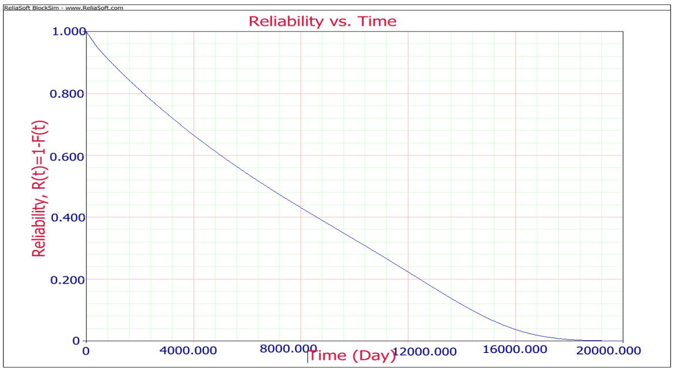

Figure 6: Reliability vs Time Graph at resolution = 50 Days

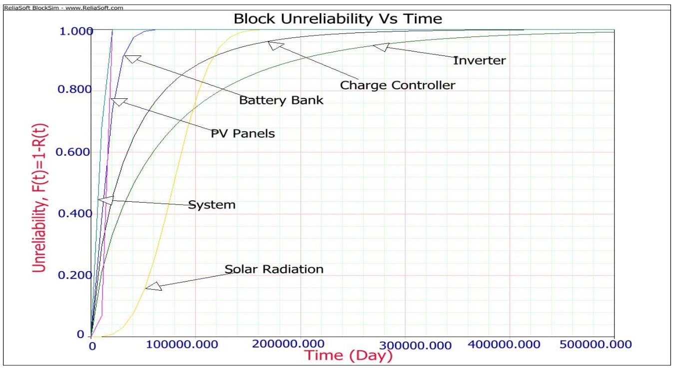

Figure 7: Block Unreliability vs. Time Graph at resolution = 50 Days

V. CONCLUSION

The research study met its aims including designing of an Off- Grid Solar PV System for a house located in Queensland (Australia) as well as modelling of the PV System and estimating the PV System's reliability with the help of ReliaSoft BlockSim 9 software. The research study discussed the factors affecting the reliability of an off grid solar PV system. The result of this research could be used to model an off grid solar PV system and estimate its reliability and availability. The research model could also be used to determine the repair rates which could help to achieve desired system availability. It was seen during the study that the reliability of an Off-Grid Solar PV System is dependent on the reliabilities of the each individual components inside the system. If a solar PV system consists of one or more unreliable components, then the overall reliability of the system would decrease. Since the failure and lifetime data of components of the Off-Grid solar PV systems were not available, therefore the data in this research was assumed and inputted into the software. The output results from the ReliaSoft BlockSim 9 software may differ if the assumed data is replaced by the actual field data. In future, the research can be carried out considering environmental factors which could impact the system at different temperatures and humidity levels. The model would then be more flexible and accurate in determining the reliability of the system in varying environments around the world. This would be accomplished via development of suitable set of accelerated testing methods. Moreover the expected operating environmental range in terms of both temperature and humidity would be determined. As some of the components of PV system are exposed to extreme temperature conditions both during the summer and the winter seasons, the future study would take into account the factors such as corrosion of components and component failures due to these factors. The future study would also incorporate the factors such as the decreasing conversion efficiency of the solar cells with the changing seasons and weather conditions and the decreasing efficiencies of the battery banks with time.

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